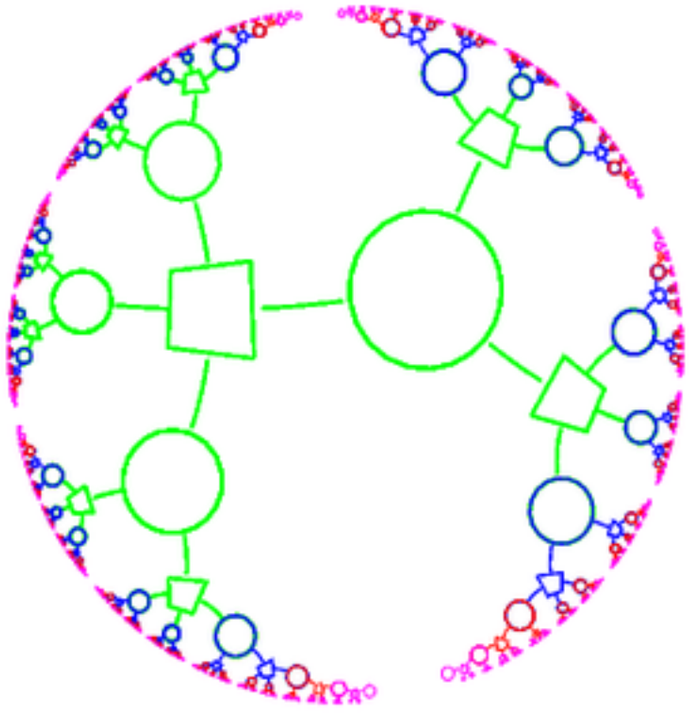
Inference & Information Measures for NOISY CHANNELS

- The Capacity



Information theory, pattern recognition, and neural networks

- 1 Noisy-channel coding

Source coding (Data compression)

- 2 Information content, entropy
- 3 Typicality and the source coding theorem
- 4 Symbol codes
- 5 Symbol codes and Arithmetic coding

Noisy-channel coding

- 6 Inference and

  Information measures for noisy channels
- 7 Capacity of a noisy channel
- 8 The noisy-channel coding theorem

Last time

- inference
- information measures for dependent variables

The rules

- Gameshow host hides prize behind one door

The rules

- Gameshow host hides prize behind one door
- Player chooses a door
  eg door 1

The rules

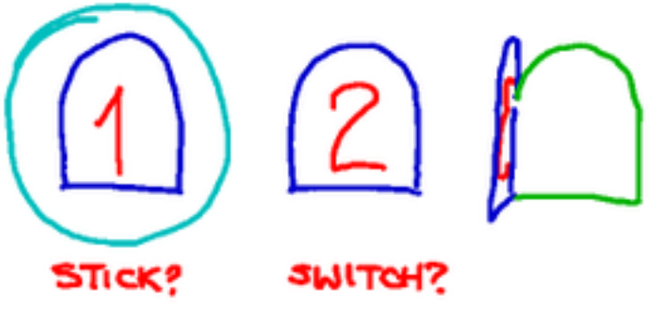
- Host opens another door, promising the prize will not be revealed.

The rules

- Host opens another door, promising the prize will not be revealed.
- Player chooses to stick or switch ie, receive what's behind his original door, or the other closed door.

Q

- Player chooses door 1
- Host opens door 3, revealing nothing, as promised



A should stick

B should switch

C makes no difference

Z don't know

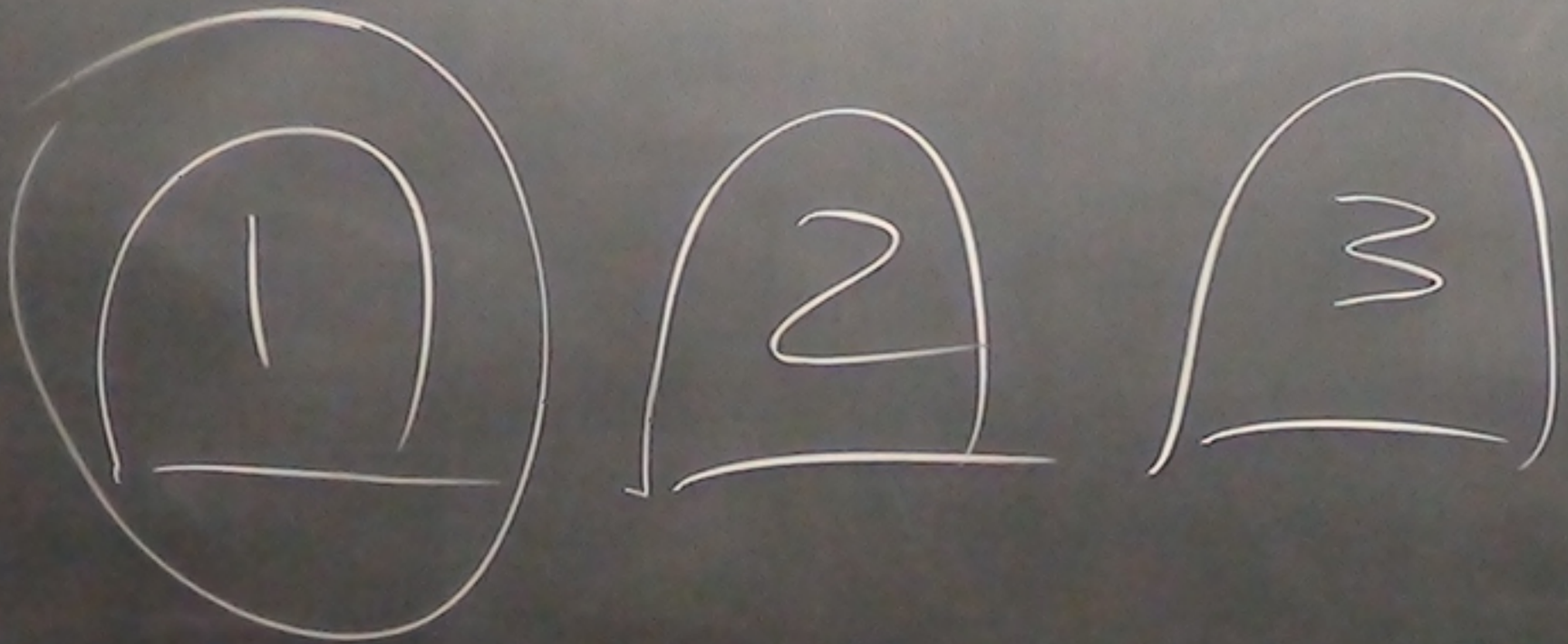
Solving Three Doors by Probability Theory

- Bayes's theorem

$$\underset{P(\mathcal{H}\,|\,\mathrm{Data})}{\overset{\text{POSTERIOR}}{}} = \frac{\overset{\text{LIKELIHOOD}}{P(\mathrm{Data}\,|\,\mathcal{H})}\,\overset{\text{PRIOR}}{P(\mathcal{H})}}{P(\mathrm{Data})}$$

HYPOTHESIS $\mathcal{H}$	PRIOR $P(\mathcal{H})$	LIKELIHOOD $P(D=3\,\|\,\mathcal{H})$	POSTERIOR $P(\mathcal{H}\,\|\,D=3)$
$\mathcal{H}=1$	$\frac{1}{3}$	$\frac{1}{2}$	
$\mathcal{H}=2$	$\frac{1}{3}$	1	
$\mathcal{H}=3$	$\frac{1}{3}$	0	

H=1 H=2 H=3



D=2 D=3

Solving Three Doors by Probability Theory

- Bayes's theorem

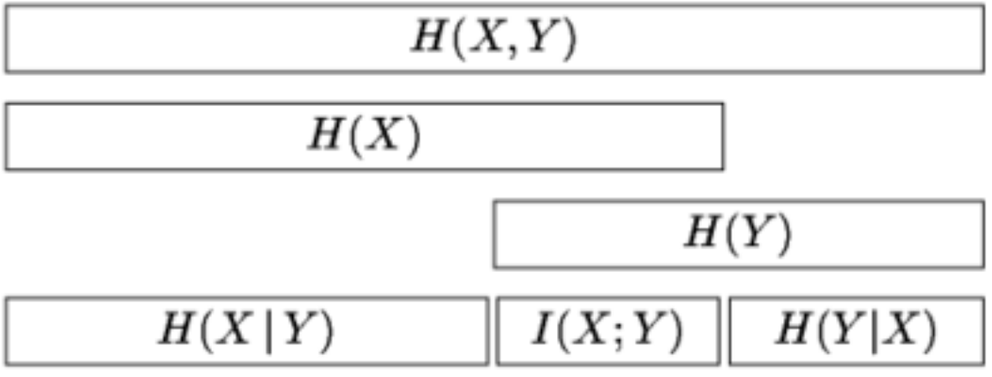
$$\overset{\text{POSTERIOR}}{P(\mathcal{H}\,|\,\text{Data})} = \frac{\overset{\text{LIKELIHOOD}}{P(\text{Data}\,|\,\mathcal{H})}\;\overset{\text{PRIOR}}{P(\mathcal{H})}}{P(\text{Data})}$$

HYPOTHESIS $\mathcal{H}$	PRIOR $P(\mathcal{H})$	LIKELIHOOD $P(D=3\,\vert\,\mathcal{H})$	POSTERIOR $P(\mathcal{H}\,\vert\,D=3)$
$\mathcal{H}=1$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}\times\frac{1}{2}/\,\frac{1}{2}=\frac{1}{3}$
$\mathcal{H}=2$	$\frac{1}{3}$	1	$\frac{1}{3}\times 1/\,\frac{1}{2}=\frac{2}{3}$
$\mathcal{H}=3$	$\frac{1}{3}$	0	0

$$P(D=3) = \sum_{\mathcal{H}} P(D=3\,|\,\mathcal{H})P(\mathcal{H})$$

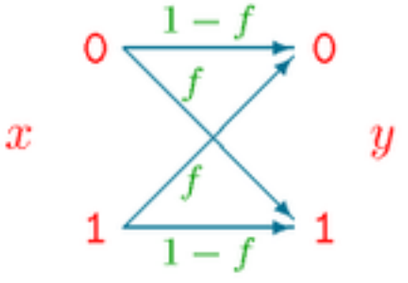
$$= \tfrac{1}{3}\times\tfrac{1}{2} + \tfrac{1}{3}\times 1 = \frac{1}{6}+\frac{1}{3} = \frac{1}{2}$$

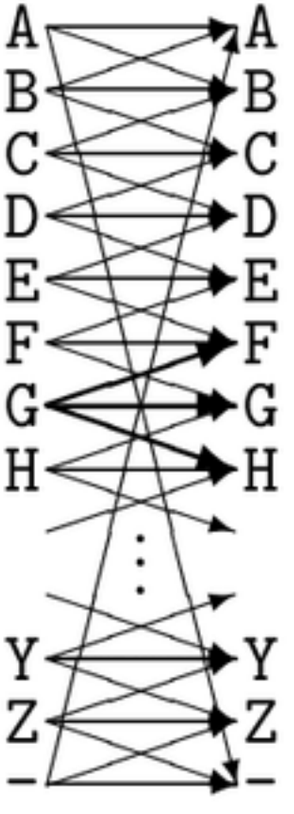
Information measures



Communication over noisy channels

- Noisy channels
  - inference
  - information measures

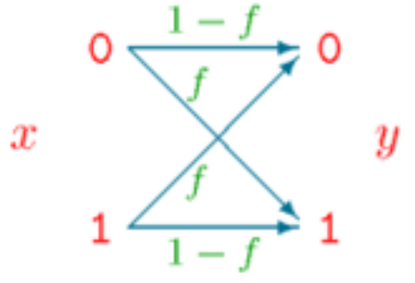




Inference for channels

Channel $\mathbf{Q}$ defines conditional probabilities

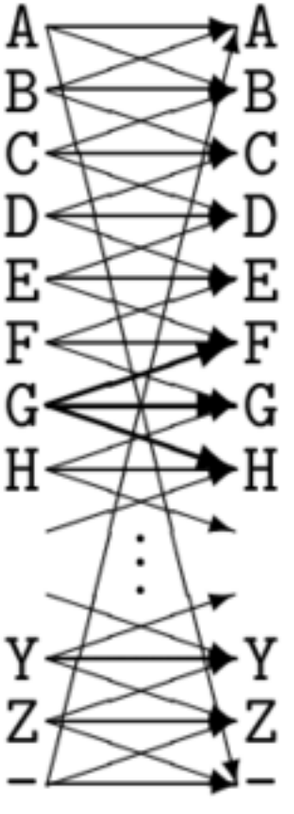
$$P(y \mid x)$$



If we choose an input distribution $P(x)$, we have a joint distribution

$$P(x, y) = P(x)\, P(y \mid x)$$

with which we can do inference.



Information measures for noisy channels

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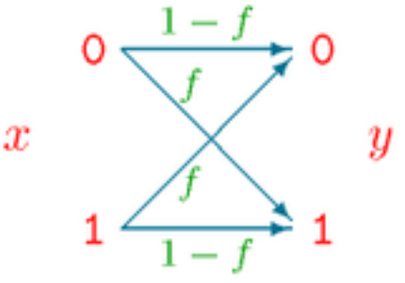
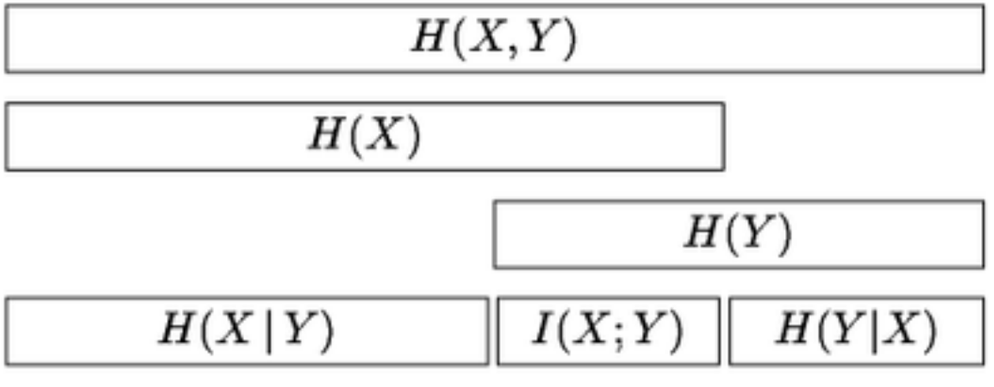


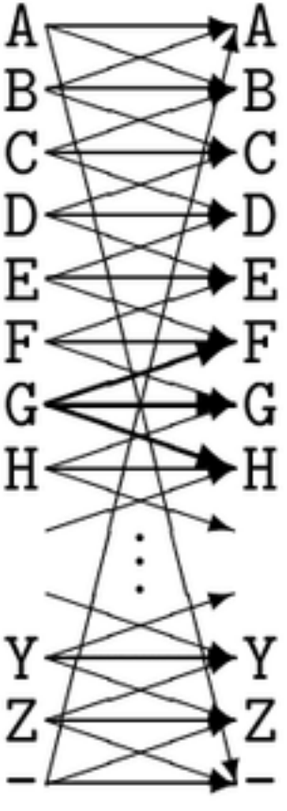
If we choose an input distribution $P(x)$,
we have a joint distribution

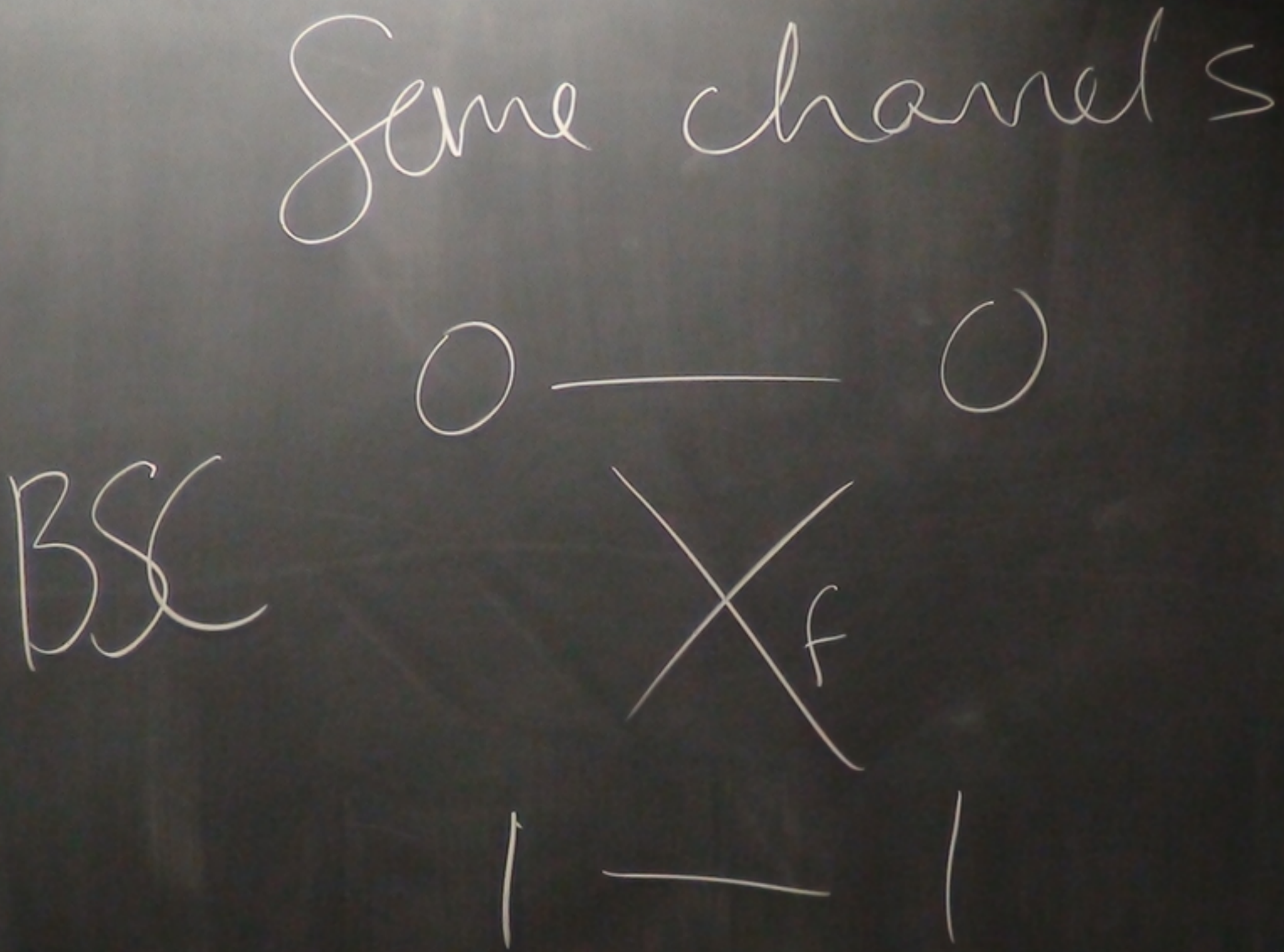
$$P(x, y) = P(x)\, P(y \mid x)$$

for which we can compute the mutual information

$$\begin{aligned} I(X;Y) &= H(X) - H(X \mid Y) \\ &= H(Y) - H(Y \mid X) \end{aligned}$$

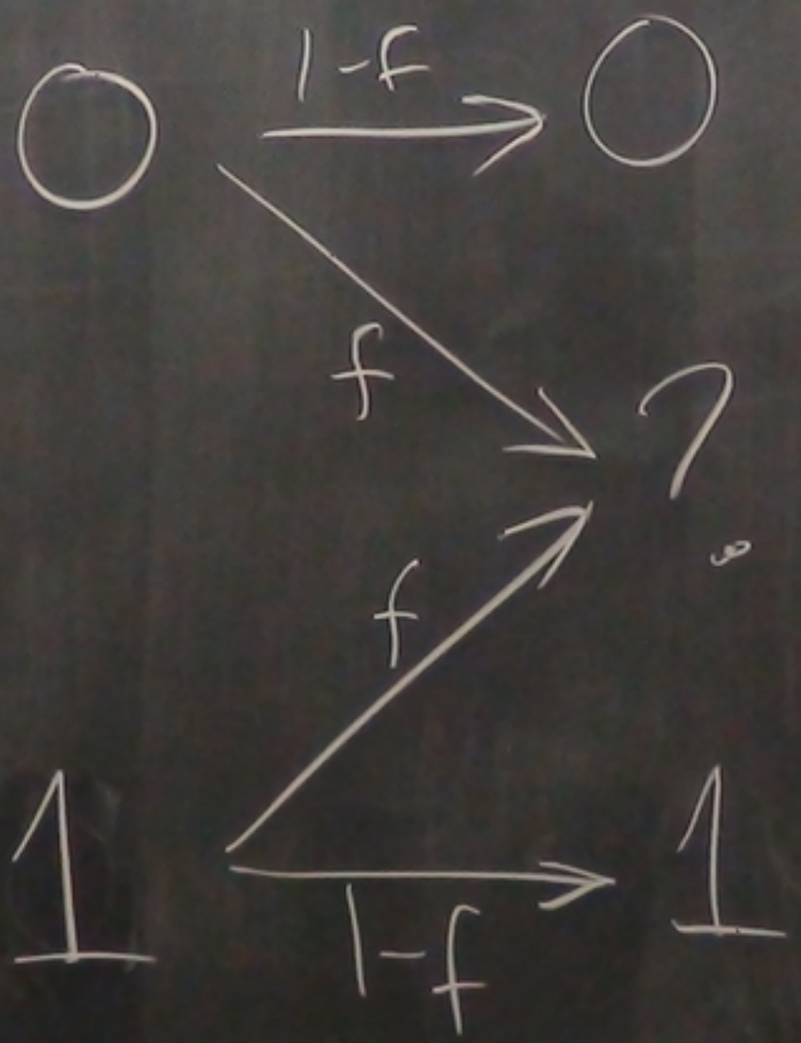





Same channels
0 — 0
BSC
f
1 — 1

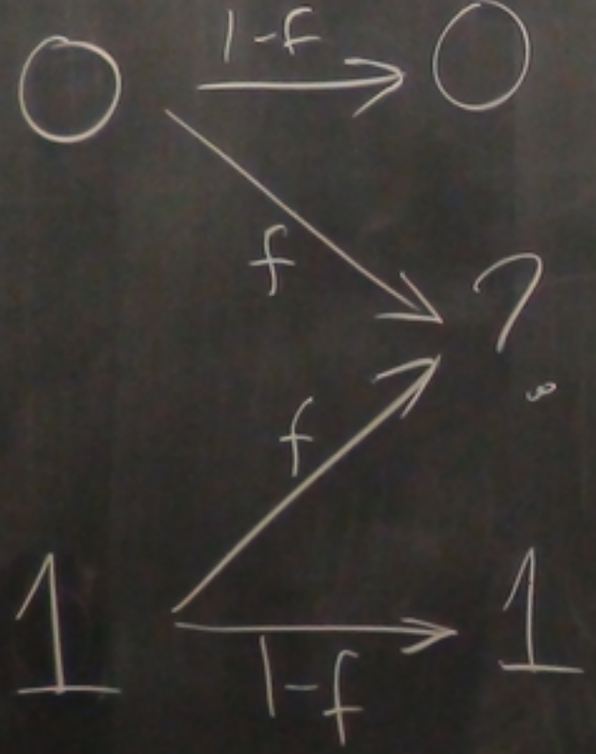
$$Q = \begin{array}{c} \\ y \\ \begin{matrix} 0 \\ 1 \end{matrix} \end{array} \begin{array}{c} \begin{matrix} x \\ 0 \qquad\quad 1 \end{matrix} \\ \begin{bmatrix} 1-f & f \\ f & 1-f \end{bmatrix} \end{array}$$

Binary erasure channel



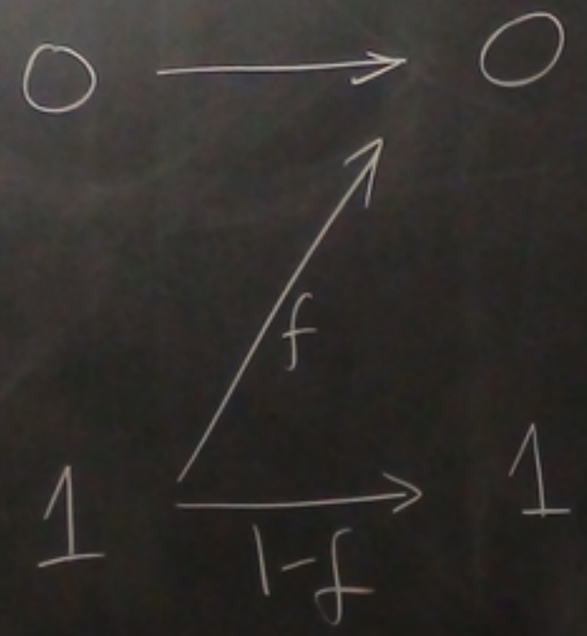
$$Q = \begin{bmatrix} 1-f & 0 \\ f & f \\ 0 & 1-f \end{bmatrix}$$

erasure channel



$$Q = \begin{bmatrix} 1-f & 0 \\ f & f \\ 0 & 1-f \end{bmatrix}$$

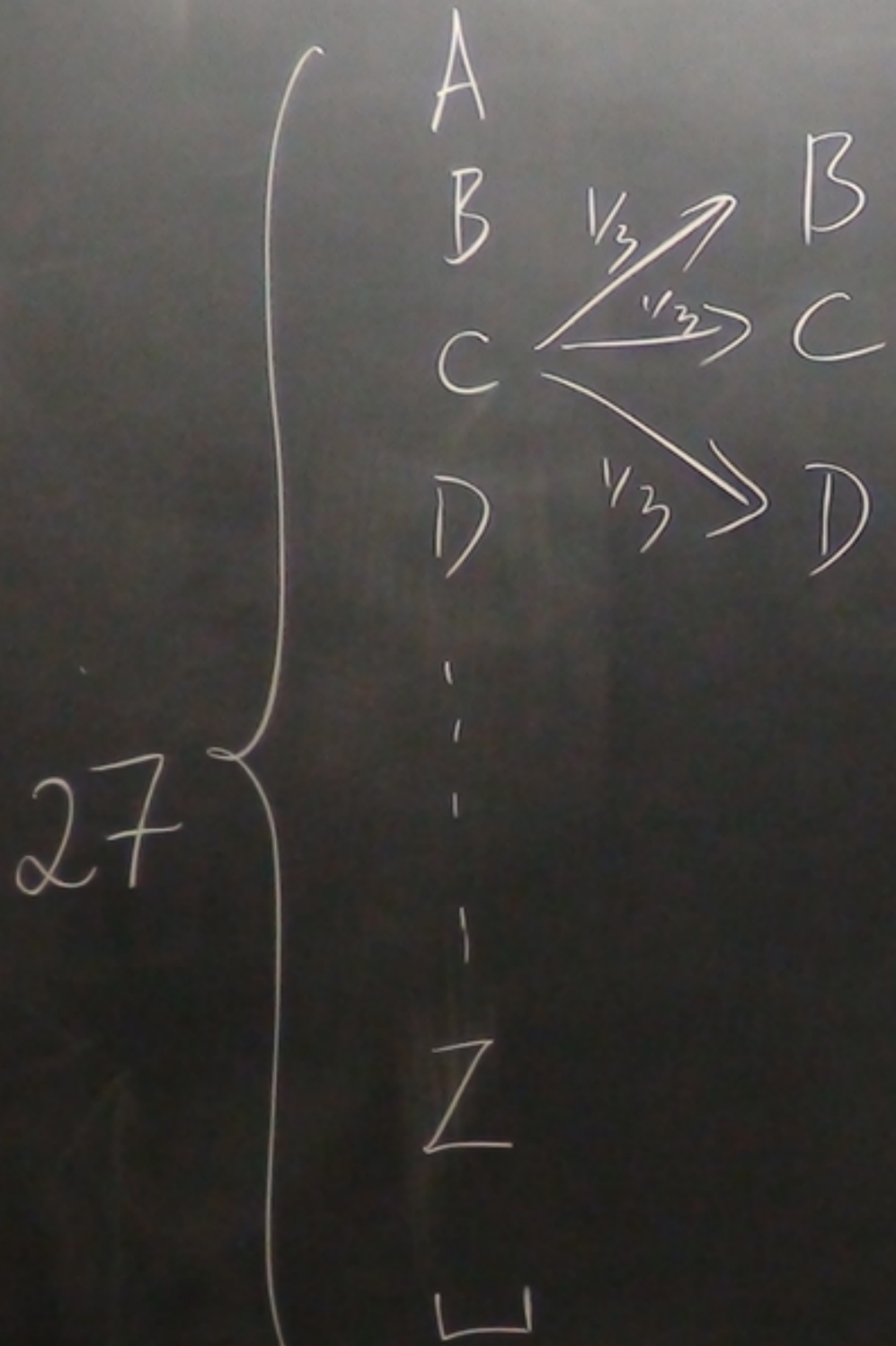
Z channel



$$Q = \begin{bmatrix} 1 & f \\ 0 & 1-f \end{bmatrix}$$

$$Q_{j|i} = P(y = b_j \mid x = a_i)$$

Nosy typewriter



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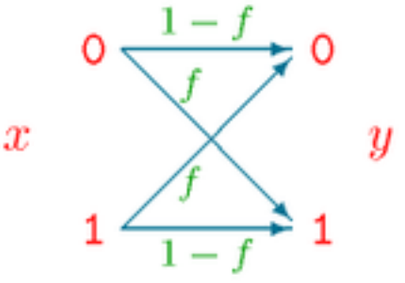

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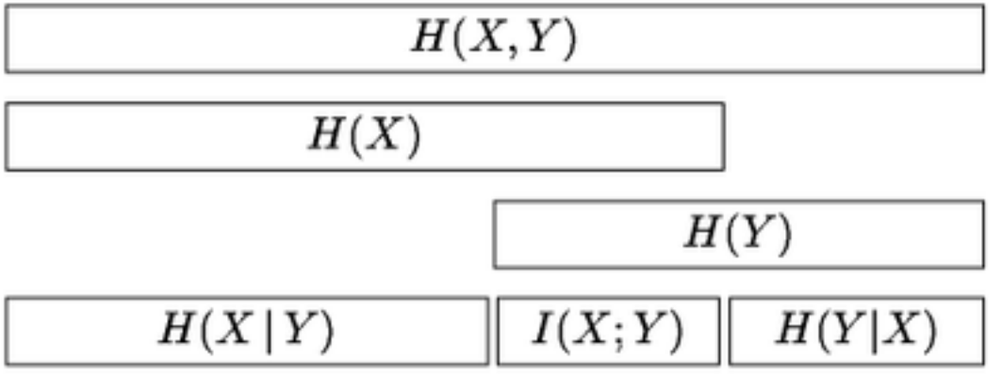
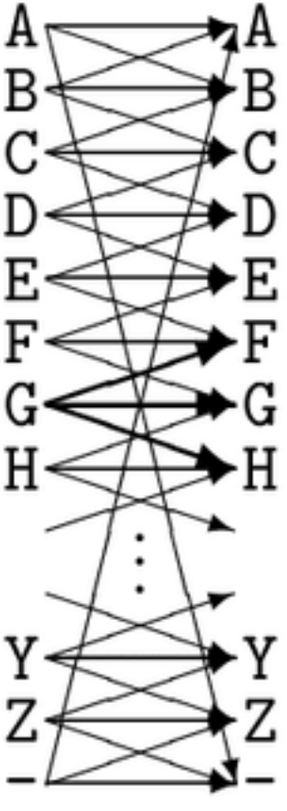
$$\begin{aligned} I(X;Y) &= H(X) - H(X \mid Y) \\ &= H(Y) - H(Y \mid X) \end{aligned}$$





Pick this

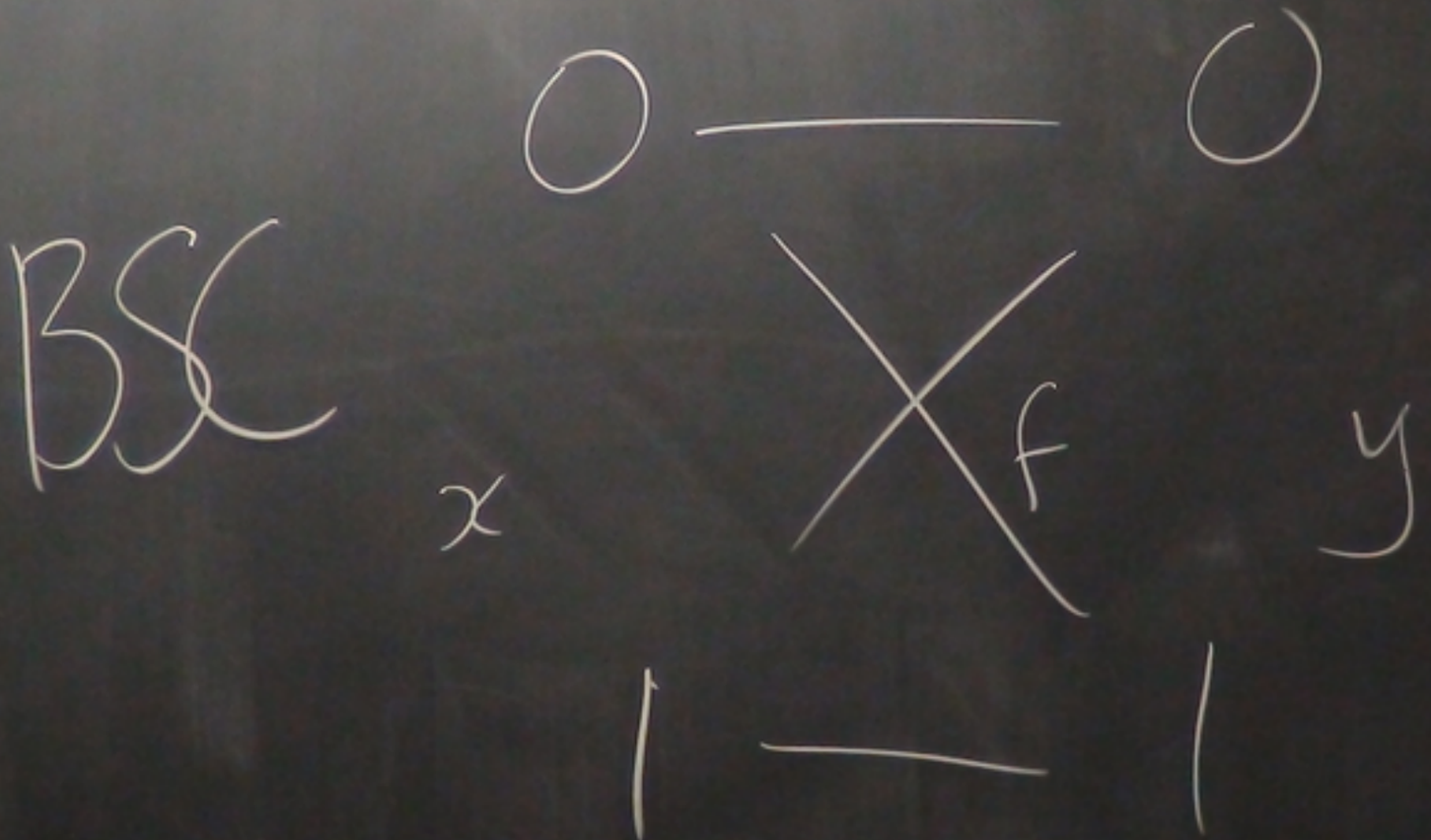
Input distbn:

$f = 0.1$

$P(x=0) = 0.9$

$P(x=1) = 0.1$

Some channels



observe $y = 1$

Given this output,

What's the probability that $x = 1$?

A	0.1	0
B	0.5	5
C	0.9	3
D	Something else	0
Z	Don't know	3

$P(x=1 \mid y$

$$P(x=1 \mid y=1) = \frac{P(y=1 \mid x=1)\, P(x=1)}{P(y=1)}$$

LIKELIHOOD PRIOR

$$P(y=1) = \sum_{x'} P(y=1 \mid x')\, P(x')$$

$$= \frac{0.9 \times 0.1}{\underbrace{0.9 \times 0.1 + 0.1 \times 0.9}_{P(y=1) = 0.18}}$$

$$P(x=1 \mid y=0) = \frac{P(y=0 \mid x=1)\, P(x=1)}{P(y=0)}$$

$$= \frac{0.1 \times 0.1}{\underbrace{0.01 + 0.81}_{P(y=0) = 0.82}} = \underline{\underline{\frac{1}{82}}}$$

LIKELIHOOD PRIOR

$$P(x=1 \mid y=1) = \frac{P(y=1 \mid x=1)\, P(x=1)}{P(y=1)}$$

$$P(y=1) = \sum_{x'} P(y=1 \mid x')\, P(x')$$

$$= \frac{0.9 \times 0.1}{0.9 \times 0.1 + 0.1 \times 0.9} = \frac{1}{2}$$

$P(y=1) = 0.18$

Info conveyed about input by the output

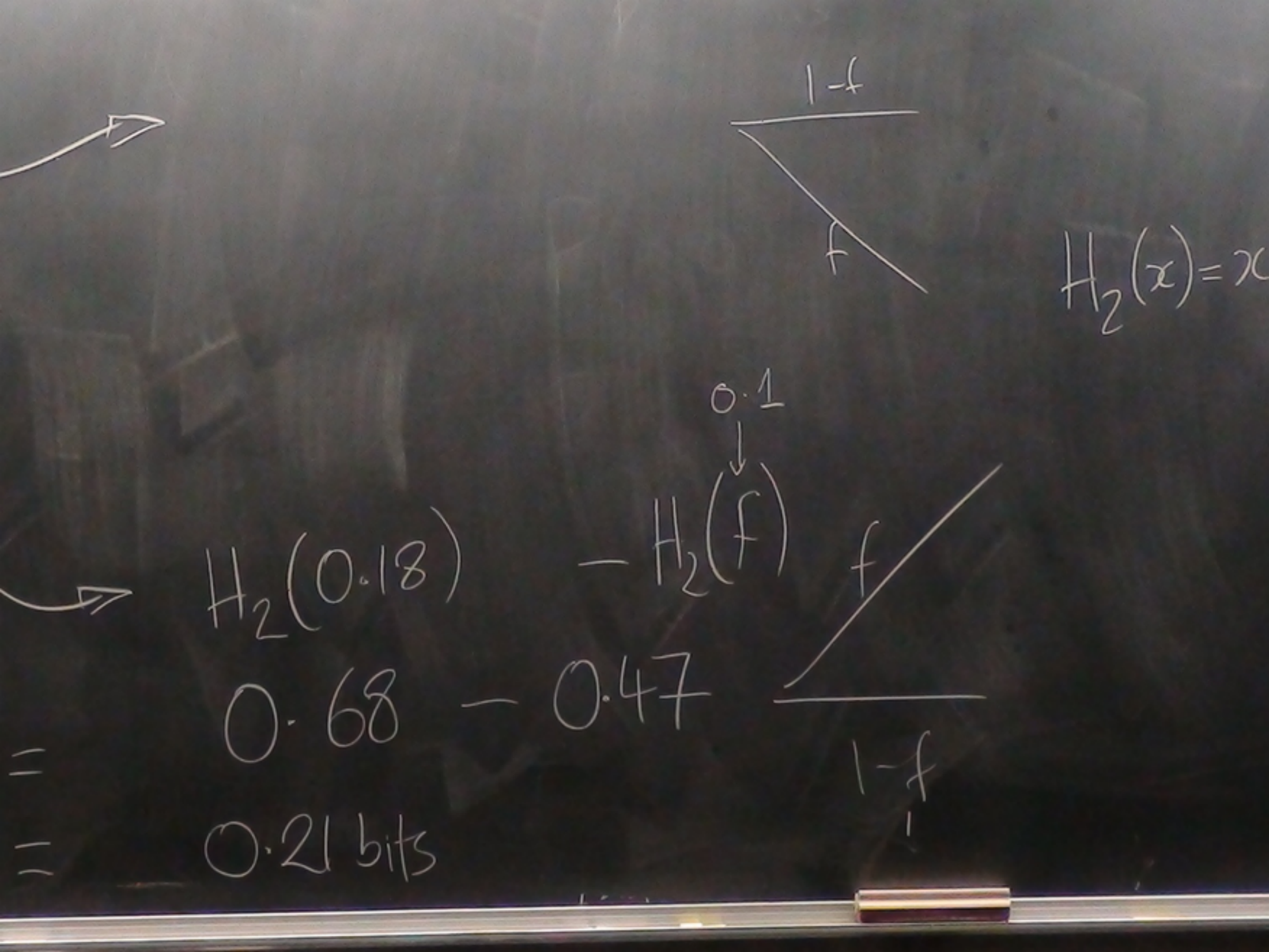
$$I(X;Y) = H(X) - H(X|Y)$$

$$= H(Y) - H(Y|X)$$

ut input by the output

$H(X) - H(X|Y)$ →

$H(Y) - H(Y|X)$ * → $H_2(0.18)$ $- H_2(f)$

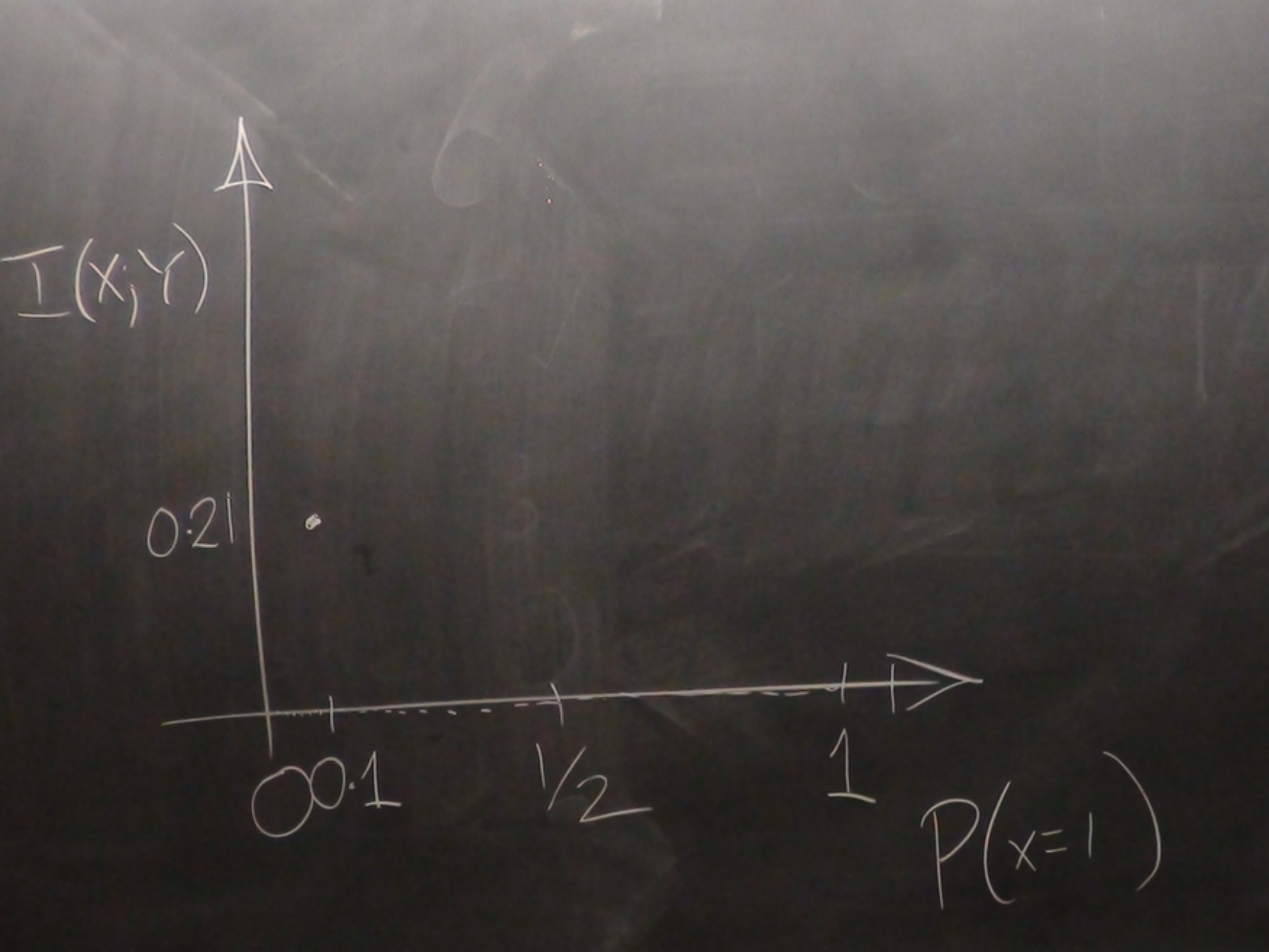

1-f
f
$H_2(x) = x$
0.1
$H_2(0.18) - H_2(f)$
$0.68 - 0.47$
f
1-f
1
=
0.21 bits
=

$$H(X) - \underbrace{H(X|Y)}_{\sum_y P(y)\, H(X|y)}$$

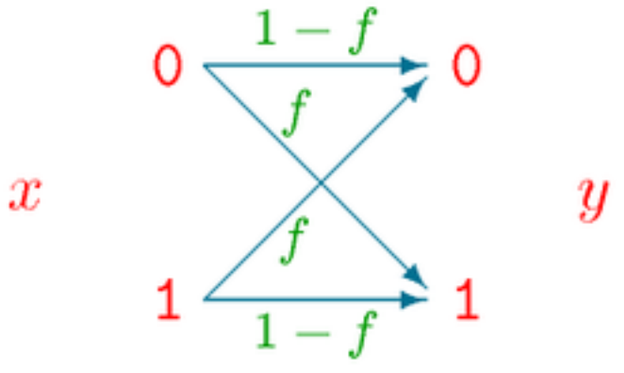
$$0.47 - \Big[\underbrace{0.18}_{(y=1)} \times H_2\Big(\overbrace{\tfrac{1}{2}}^{\text{BT}}\Big) + \underbrace{0.82}_{(y=0)} \times H_2\Big(\overbrace{\tfrac{1}{82}}^{\text{BT}}\Big)\Big]$$

$$= 0.47 - 0.26$$

$$= \underline{\underline{0.21}} \text{ bits}$$


I(X;Y)
0.21
0
0.1
1/2
1
P(x=1)

Mutual information for the BSC



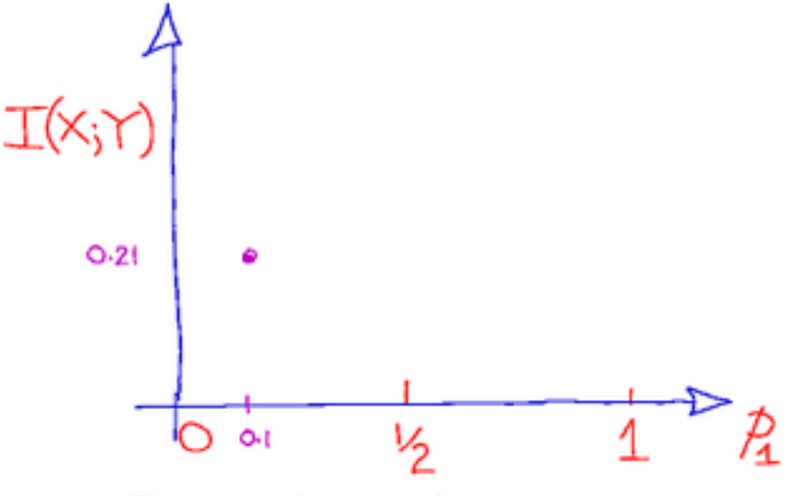
$$\mathbf{Q} = \begin{bmatrix} 1-f & f \\ f & 1-f \end{bmatrix}$$

Consider $f = 0.1$.

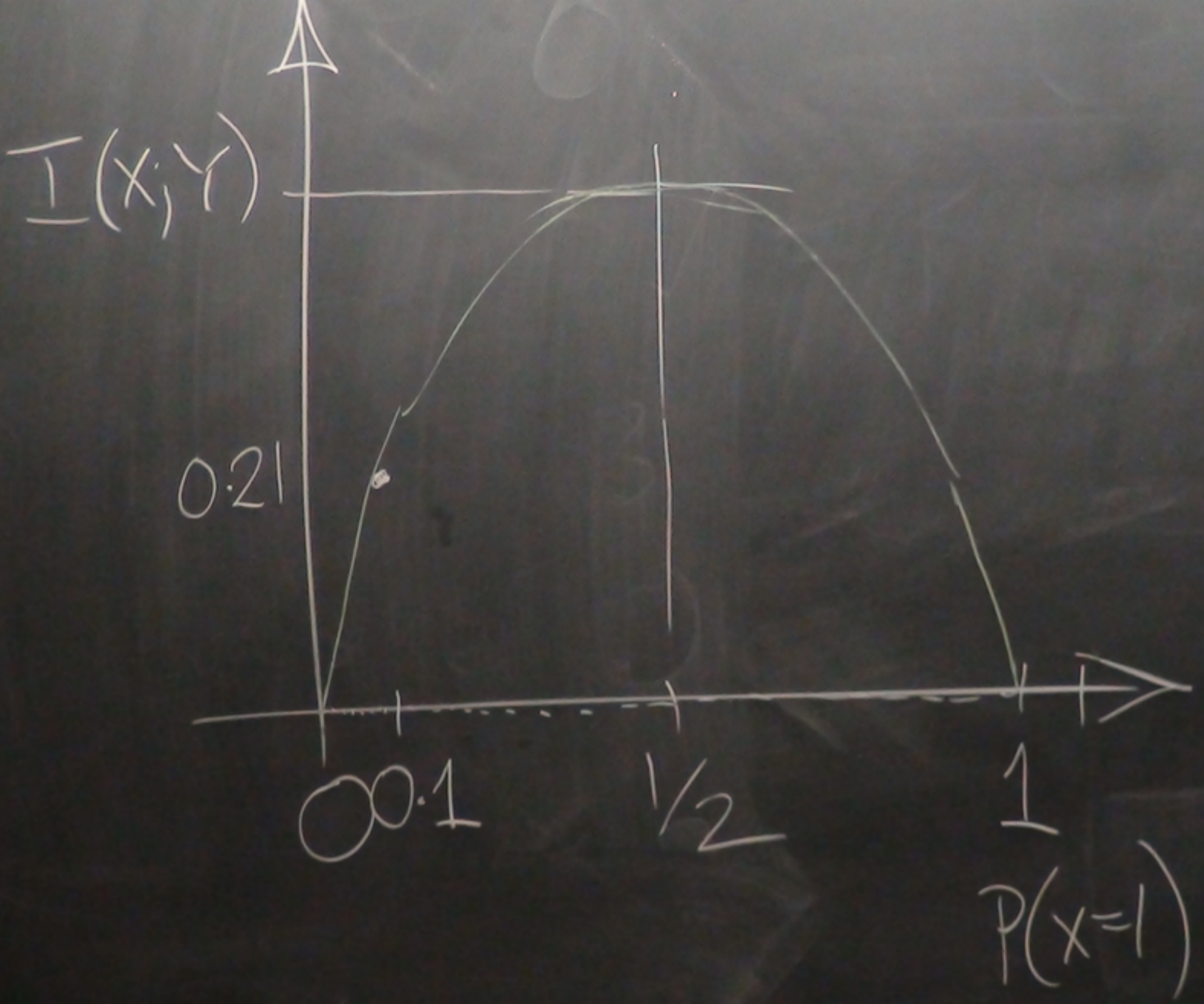
Assume input distribution $\mathcal{P}_X = \{p_0, p_1\} = \{0.9, 0.1\}$.

Find

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y \mid X) \\ &= H_2(0.18) - H_2(0.1) \\ &= 0.68 - 0.47 \\ &= 0.21 \text{ bits.} \end{aligned}$$



Complete the curve


I(X;Y)
0.21
0
0.1
1/2
1
P(X=1)

Definition

Capacity of a channel

$$C(Q) = \max_{P_X} I(X;Y)$$

The Capacity of a channel

is the maximum, over all input distributions $P(x)$, of the mutual information:

$$C \equiv \max_{\mathcal{P}_X} I(X;Y)$$

The distribution $\mathcal{P}_X^*$ that achieves the maximum is called the optimal input distribution.

Definition

Capacity of a channel

$$C(Q) = \max_{P_X} I(X;Y)$$

eg for BSC with $f = 0.1$, the opt. inp. dist is $\left(\frac{1}{2}, \frac{1}{2}\right)$

input distbn $\rightarrow P_1 \qquad (P_1, 1-P_1)$

$$I(X;Y) = H(Y) - H(Y|X)$$

$$= H_2\Big(P_1(1-f) + (1-P_1)\times f\Big)$$

$$- H_2(f)$$

Maximize

$$C = 1 - H_2(f)$$

$-p_i$)

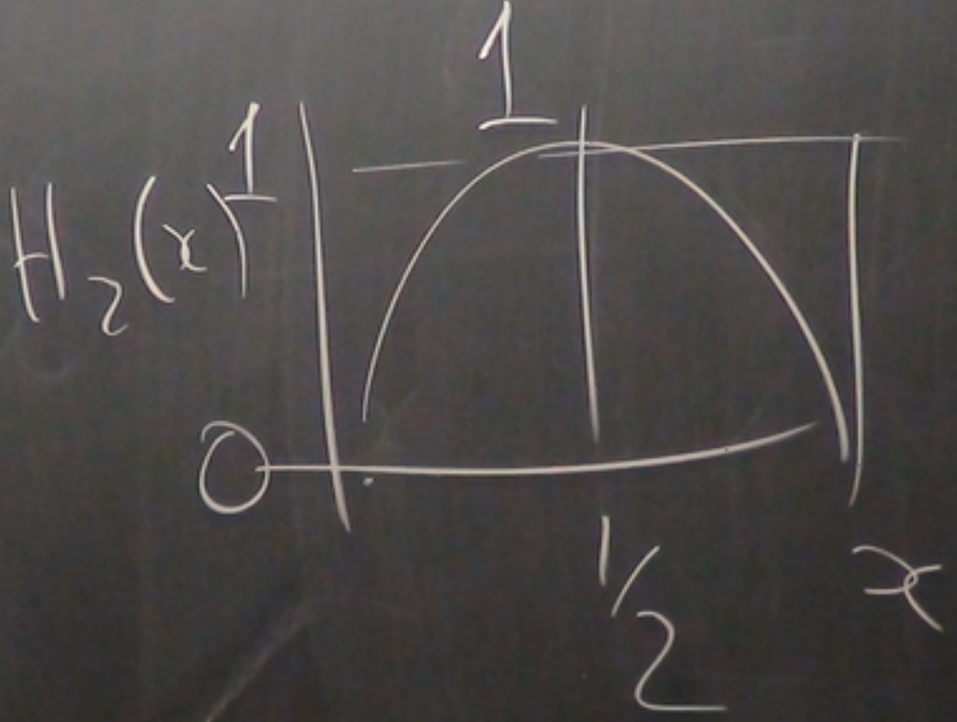
$+ (1-p_i) \times f$)

f)

$_2(f)$

Info

$I(X;Y)$



The Capacity of a channel

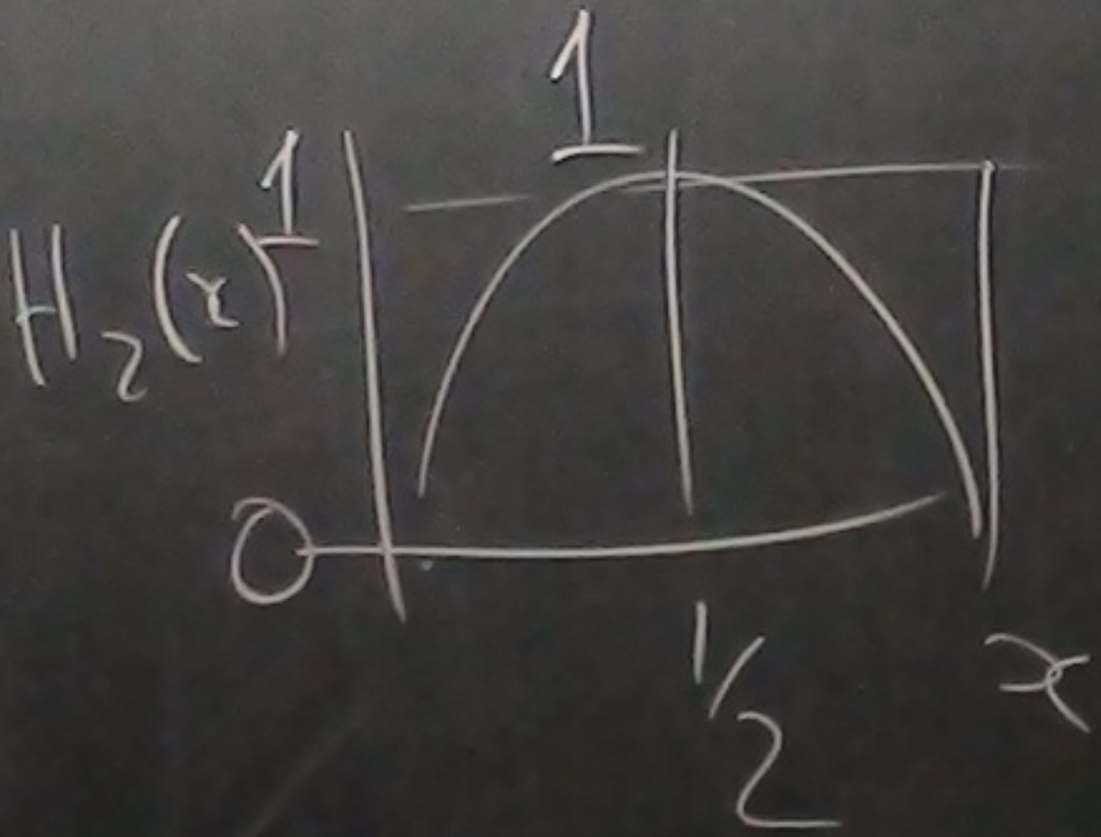
is the maximum, over all input distributions $P(x)$, of the mutual information:

$$C \equiv \max_{\mathcal{P}_X} I(X;Y)$$

The distribution $\mathcal{P}_X^*$ that achieves the maximum is called the optimal input distribution.

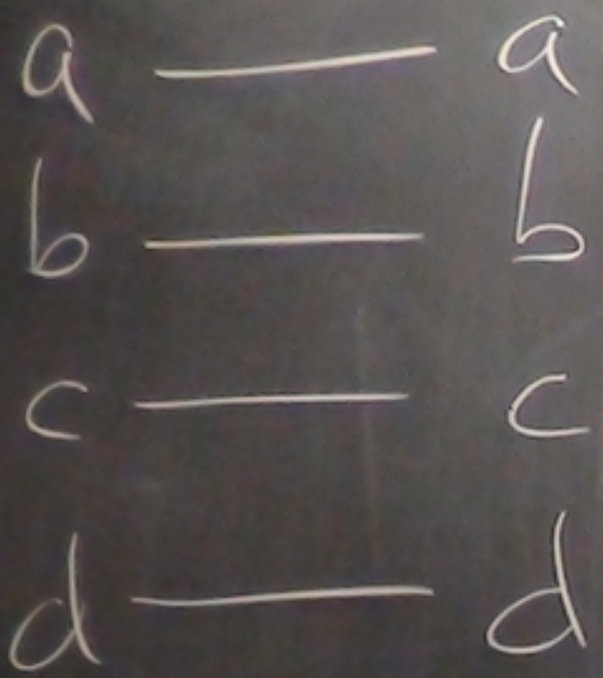
- Shannon's noisy-channel coding theorem:

Reliable (virtually error-free) communication is possible at rates up to C



$$C_{BSC} \simeq 0.53$$

$f = 0.1$



$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

What is the optimal input distribution?

$$p_X^* = \left\{ \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4} \right\}$$

$$C = 2 \text{ bits}$$

$$= H(Y) - H(Y|X)$$

$$= \log_2 4 - 0 \text{ bits}$$

$$= 2 \text{ bits}$$

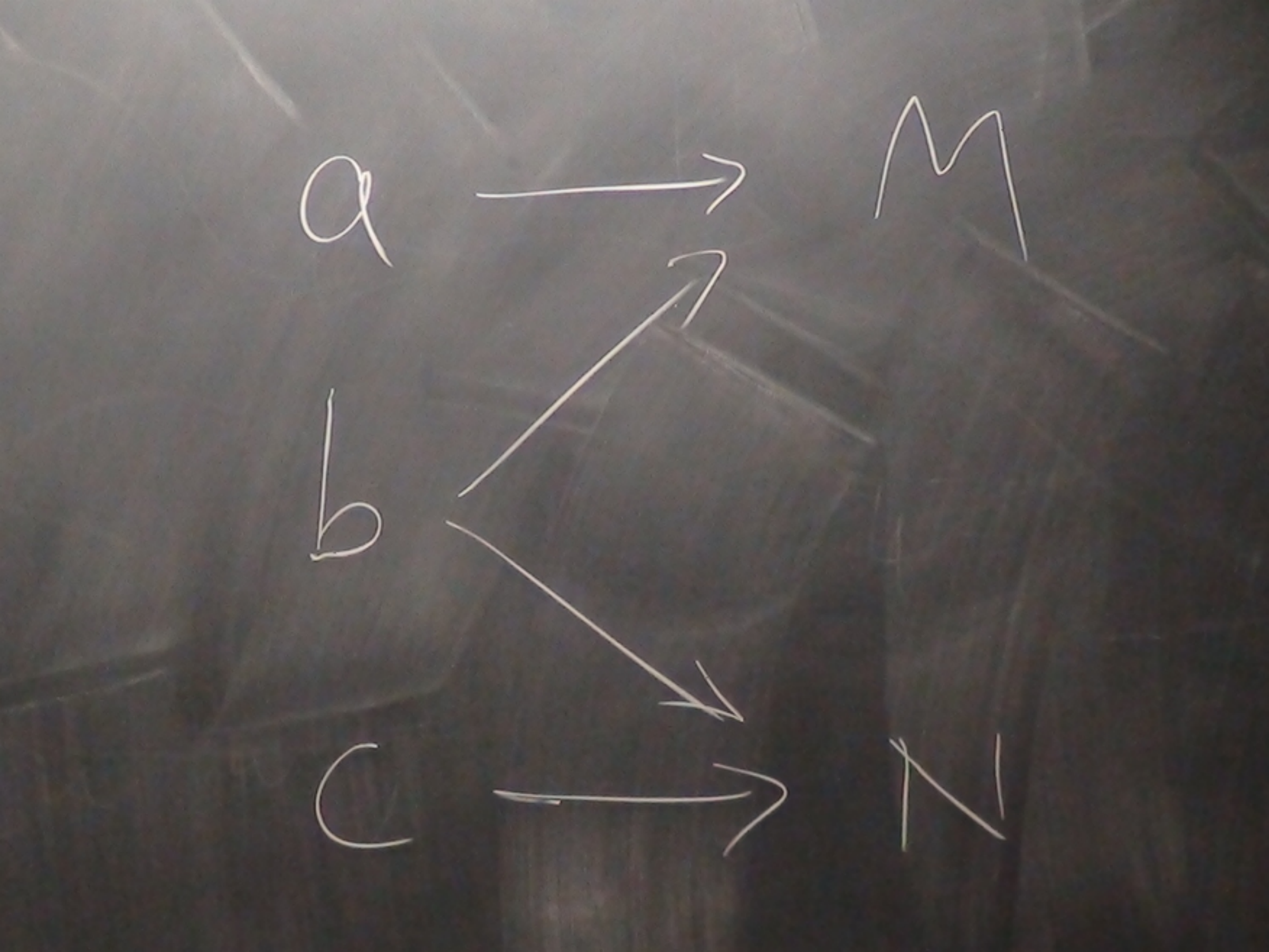
00 a — a

01 b — b

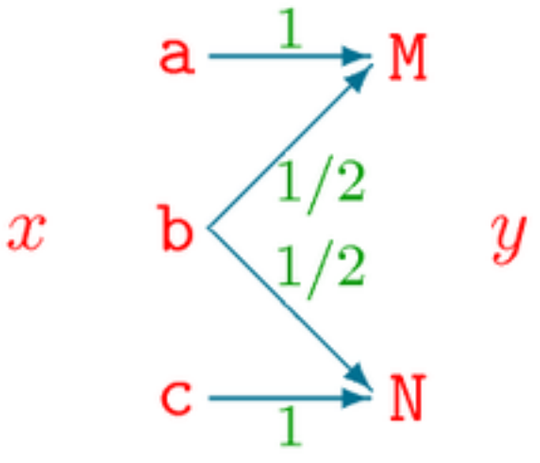
10 c — c

11 d — d

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$


a
M
b
c
N

Ternary confusion channel



$$\mathbf{Q} = \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1/2 & 1 \end{bmatrix}$$

Assume input distribution $\mathcal{P}_X = \left\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right\}$.

- If $y = M$, what's x?
- What is the mutual information $I(X;Y)$?

Assume $P_X = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$

If $y = M$, what's x?

$$P(x \mid y = M) = \begin{cases} 2/3 \\ 1/3 \\ 0 \end{cases}$$

What's $I(X;Y)$

$$I(X;Y) = H(X) - H(X|Y)$$

ω	Something else
A	$\log 3 - H_2(1/3)$
B	$1 - 1/3 = 2/3$
C	$\log 3 - 1/3$
D	$\log 3 - 1$
E	$\frac{1}{2} \log 3$
Z	Don't know

ω	Something else		
✓ A	$\log_2 3 - H_2(1/3)$	✓	0 ✓
✓ B	$1 - 1/3 = 2/3$		4 ✓
C	$\log 3 - 1/3$		0
D	$\log 3 - 1$		1
E	$\frac{1}{2} \log 3$		0
Z	Don't know		6

$$X|Y) = H(Y) - H(Y|X)$$

$$\frac{1}{3}\times 0 + \frac{1}{3}\times H_2\left(\frac{1}{2}\right) \quad \frac{1}{3}\times 0$$

$$\frac{1}{3}\times 1$$

$$= 1/3$$

✓ 0 ✓

✓ 4

0

1

0

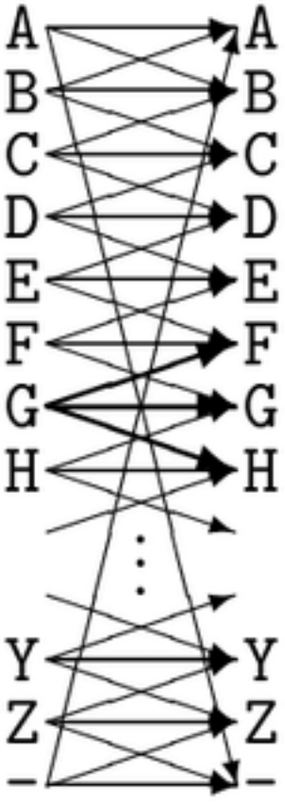
6

Ternary confusion channel

$$P_X^* = \left\{ \frac{1}{2}, 0, \frac{1}{2} \right\}$$

& Capacity = 1 bit

Noisy typewriter



$$
\begin{aligned}
&\vdots \\
P(y=\mathtt{F}\,|\,x=\mathtt{G}) &= 1/3; \\
P(y=\mathtt{G}\,|\,x=\mathtt{G}) &= 1/3; \\
P(y=\mathtt{H}\,|\,x=\mathtt{G}) &= 1/3; \\
&\vdots
\end{aligned}
$$

- What is the optimal input distribution?
- What is the capacity?

Recommended homework

- Reading: Chapters 1-6; Advance reading: Chapters 8, 9, 10
- Arithmetic coding: 6.17 (p125) 6.3 (p118) 6.7 (p123)
- Data compression recap:
  - 5.26 (p103)
  - Invent a question about data compression.
- See handout 2 on website for more recommended exercises
  - Huffman programs huffman.p, huffman.py are on website
  - also a 'bent coin' file 0010000.... as a compression benchmark